

# Keep MERLIN in control; let AI fail without stopping a truck

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- External, asynchronous, fail-open AI sidecar
- Statistical forecast ML + deterministic replenishment rules
- MERLIN generates every order and alone sends supplier EDI
- No valid AI output = unchanged min/max replenishment
- Replay and shadow only until measured release gates pass

# Change only the edges: three supported MERLIN extensions



## Can change

External sidecar  
Narrow adapters  
Control + immutable audit



## Cannot change

MERLIN engine + authority  
POS and fixed EDI schema  
04:00 supplier cut-off



## Unsafe to touch

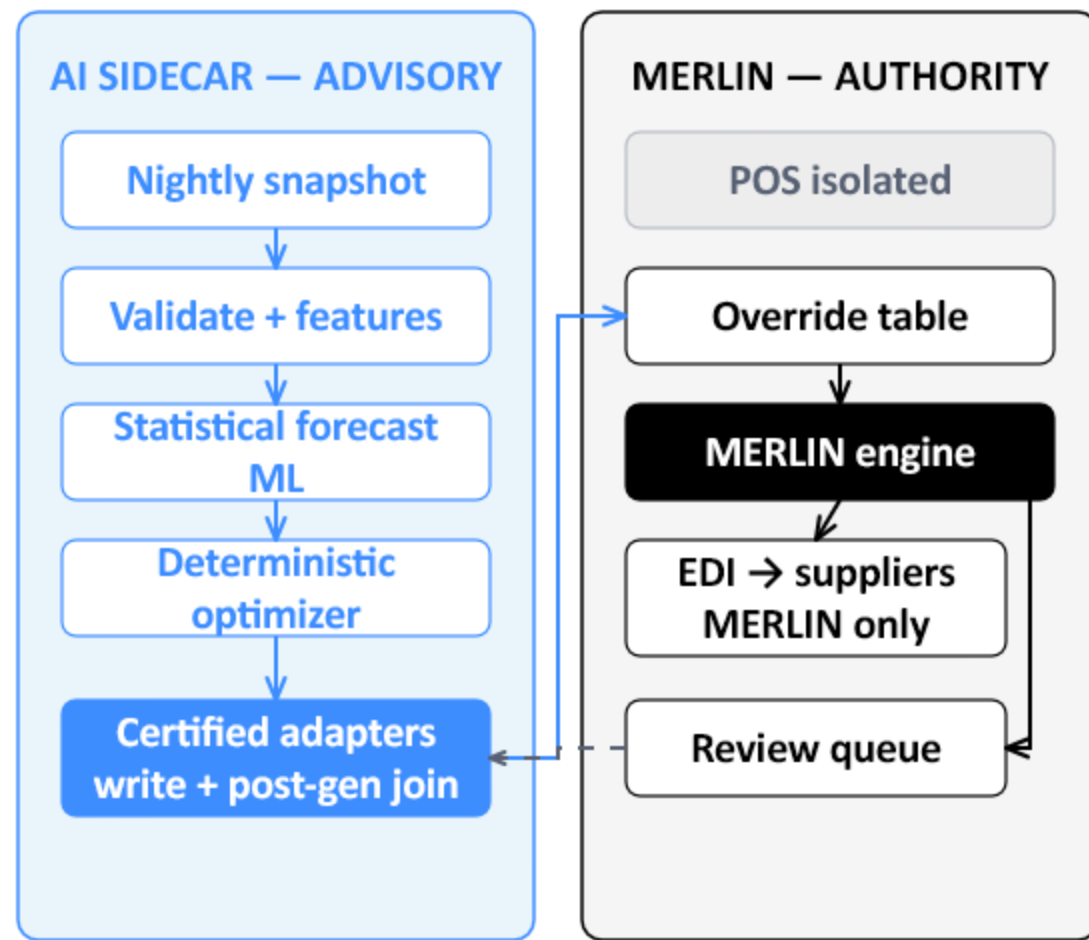
Scheduler + undocumented tables  
Manual rows  
Point inventory as truth

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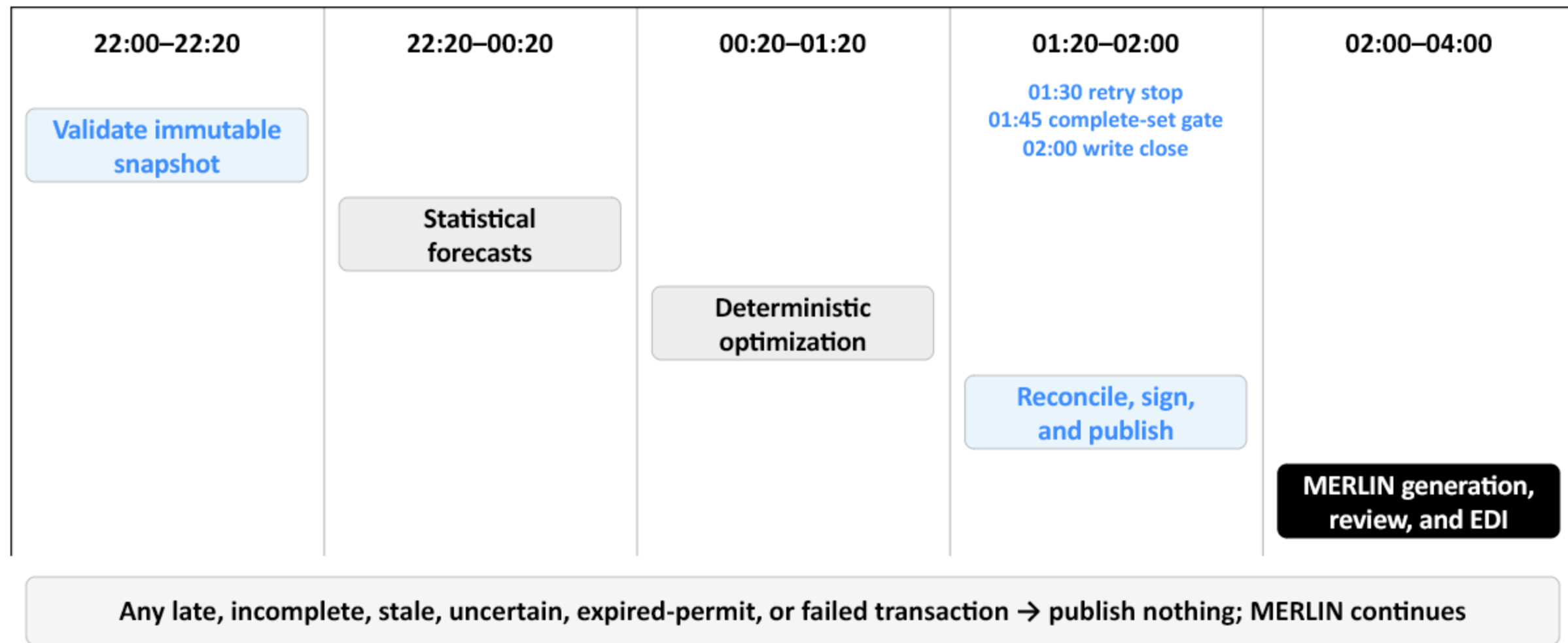
Only supported crossings → nightly batch • pre-generation override table • post-generation review queue

# The sidecar is advisory; authority stays inside MERLIN

- Nightly snapshot → validation → features → statistical forecast ML → deterministic optimizer
- Only certified adapters can write bounded current-run rows
- MERLIN generates orders; current quantity returns only after a certified post-generation join
- Loyalty PII stays country-local; copilot remains separate from POS and ordering



# AI closes at 02:00 to protect the immovable 04:00 cut-off



# The envelope fits on paper; production waits for proof

Publication proof: 70,400 rows / 60 s requires  $\geq 1,173.4$  rows/s raw; 1,760 rows/s with 10 s overhead and 20% headroom

Release gates: three consecutive 6x passes • MERLIN P99  $\leq 90$  min • certified adapter • observed bills + Finance approval

**27M / 7.04M**

Forecasts / stocked candidates  
6x test: 162M / 42.24M

**198.7 / 240**

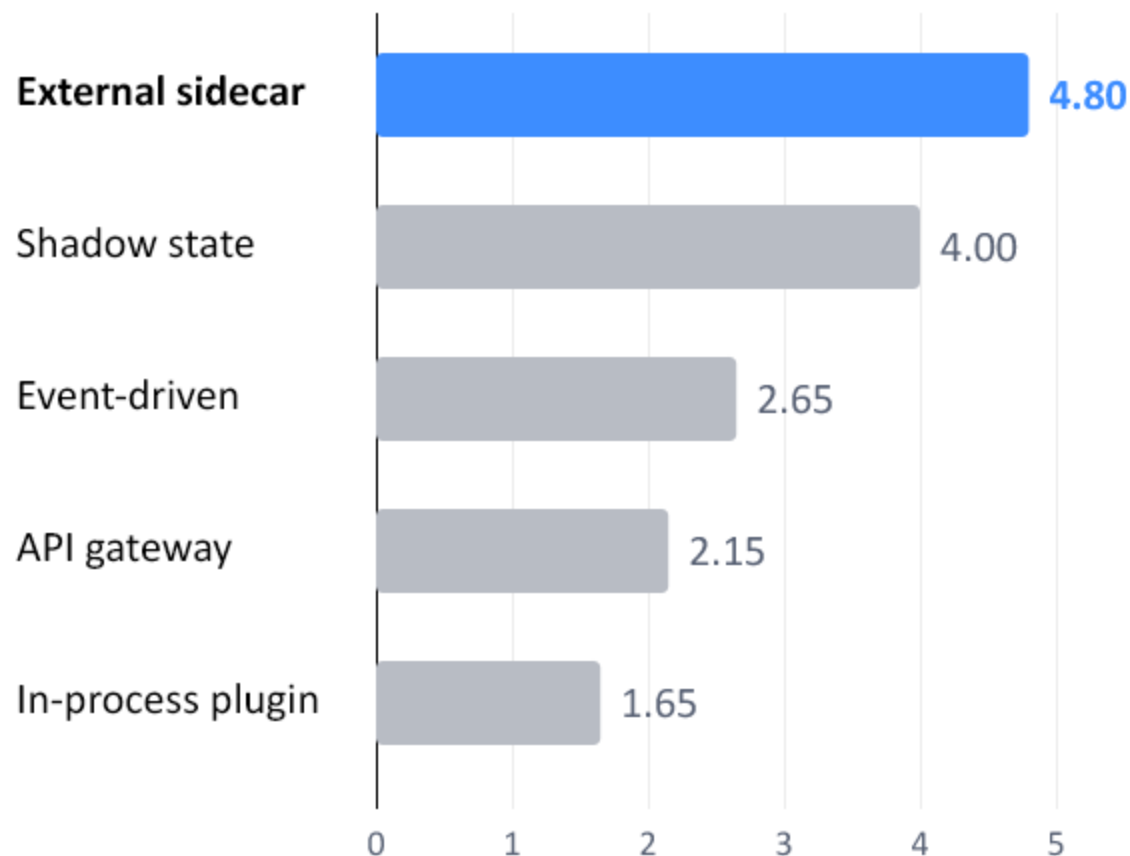
Minutes on target path  
41.3 min sidecar headroom

**\$3.8M / \$4M**

Annual target / ceiling  
\$0.0003856 per nominal forecast

TARGETS, NOT BENCHMARK RESULTS

# Sidecar wins because isolation outweighs convenience



## Decision

Use only nightly batch, override table, and post-generation review queue.

## Accepted trade-off

Duplicate snapshots, reconcile divergence, and operate two systems.

## Evidence boundary

Scores choose direction; measured release gates decide influence.

# Every top risk has a threshold and a safe fallback

All five exceed tolerance for unbounded influence; every stop condition leaves MERLIN and POS independent.

| Risk / score      | Control                               | Critical trigger                                | Safe response               |
|-------------------|---------------------------------------|-------------------------------------------------|-----------------------------|
| Deadline 20       | 01:30 stop; 01:45 gate; 02:00 close   | Reserve <30 min or EDI $\geq$ 04:00             | Bypass AI                   |
| Inventory 20      | Calibrated interval + absolute caps   | Coverage <85% or missing interval               | Suppress row/scope          |
| Peak / model 16   | 3x 6x tests + slice gates             | WAPE >30% or forecast <22,500/s                 | Disable slice               |
| Legacy adapter 15 | Epoch + permit + exact ledger         | Stale commit, manual-row change, MERLIN wait >0 | Kill / shadow               |
| Safety / legal 15 | Exact approved field + fixed template | Unsupported output or universal refusal         | Refuse / disable / escalate |

Owner: TRẦN QUỐC THUẬN (112287)

# Earn influence; disable it in minutes

## Discovery + replay

Certify contracts and timing  
Prove quality, 6x capacity, cost,  
and failure drills

## Shadow $\geq 4$ weeks

Zero MERLIN writes  
Representative promotions  
Promotion evidence required

## Pilot $\rightarrow$ scale

20 stores; 2 low-risk categories  
 $\leq \min(5,000, \text{measured cap})$  rows  
 $\leq 100$  reviews / night

deny-default epoch  $\rightarrow$  short permit  $\rightarrow$  independent SAP revoke  $\rightarrow$  exact-row cleanup

Certify

Prove

Bound + scale

Kill proof: acknowledge  $\leq 30$  s • reject new writes  $\leq 60$  s • verified safe  $\leq 5$  min — Remove the sidecar and MERLIN still orders.